전북지역혁신플랫폼
Jeonbuk Regional Innovation Platform

JST

지능형모빌리티 설계

4주차 ROS2 프로그래밍 #2

2차시 ROS2 Service



* 본 영상은 무단 배포할 수 없습니다.

학습내용

1 Service 기반 ROS 구성

2 Service 사용 방법

학습목표

- Service 사용 실습을 통해 ROS 시스템을 만들어볼 수 있다.
- Service 결과를 확인할 수 있다.

Service 기반 ROS 구성

Custom Service 생성

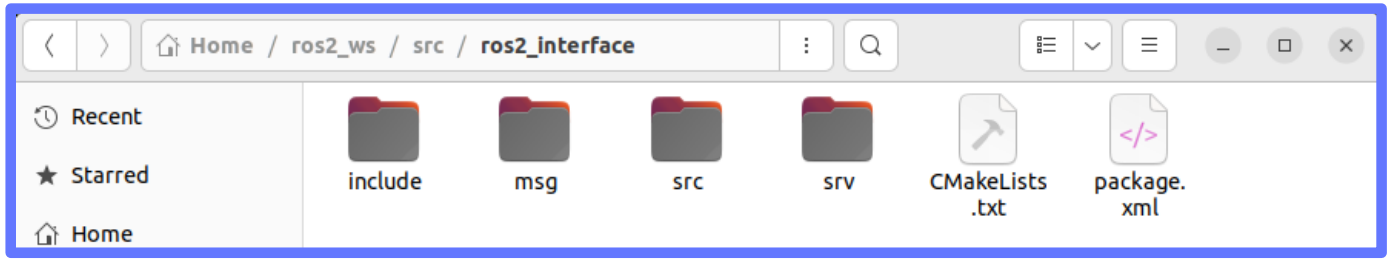
패키지 생성

- Custom Service/Message는 다른 패키지로 생성해 관리하는 것을 권장

```
$ cd ~/ros2_ws/src
$ ros2 pkg create --build-type ament_cmake ros2_interface
```

- srv: Custom Service 디렉토리(폴더)
- msg: Custom Message 디렉토리(폴더)

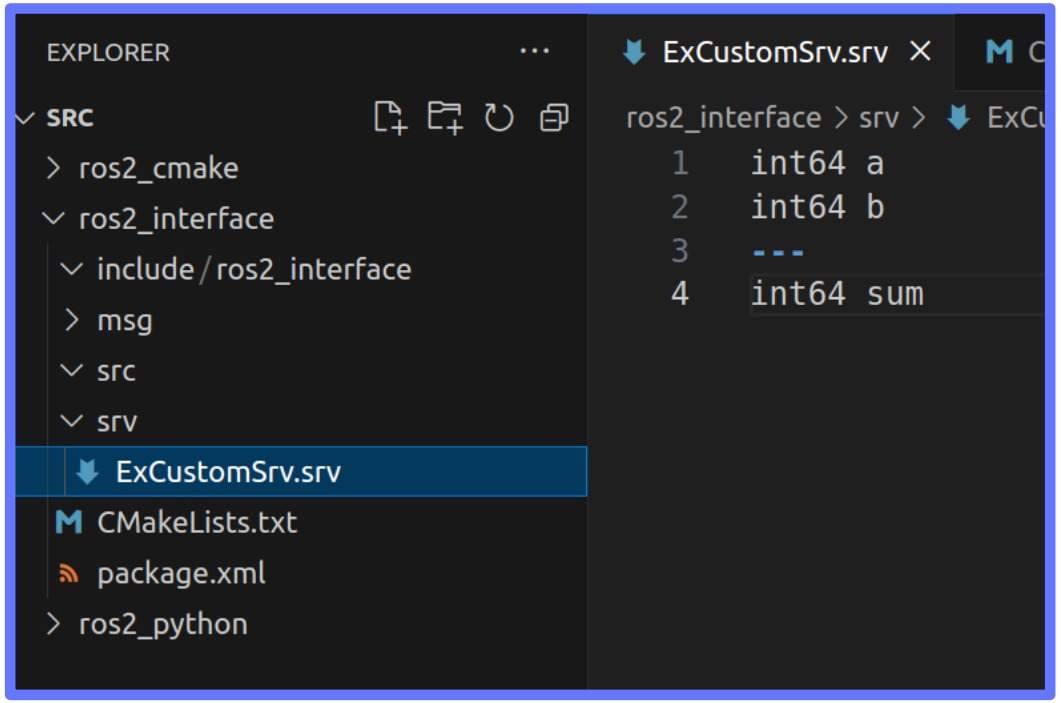
```
$ cd ros2_interface
$ mkdir srv
$ mkdir msg
```



Custom Service

srv

```
$ cd srv
$ code ExCustomSrv.srv
```



Request
- - -
Response

Custom Service

CMakeLists.txt

```
find_package(rosidl_default_generators REQUIRED)
rosidl_generate_interfaces(${PROJECT_NAME}
"srv/ExCustomSrv.srv"
DEPENDENCIES
)
```

srv 파일

Custom Service

package.xml

```
...
<build_depend>rosidl_default_generators</build_depend>
<exec_depend>rosidl_default_runtime</exec_depend>
<member_of_group>rosidl_interface_packages</member_of_group>
...
```

Custom Service

Build

```
$ cd ~/ros2_ws/
$ colcon build --packages-select ros2_interface
$ source install/setup.bash
```

Custom Service

생성한 Service 확인

```
$ ros2 interface show ros2_interface/srv/ExCustomSrv
```

출력

```
int64 a
int64 b
---
int64 sum
```

Service 사용 방법

Service - cpp

service_server.cpp

```
$ cd ~/ros2_ws/src/ros2_cmake/src/
$ code service_server.cpp
```

```cpp
#include "rclcpp/rclcpp.hpp"
#include "ros2_interface/srv/ex_custom_srv.hpp"
#include <memory>
void add(const std::shared_ptr<ros2_interface::srv::ExCustomSrv::Request> request,
         std::shared_ptr<ros2_interface::srv::ExCustomSrv::Response>      response)
{
 response->sum = request->a + request->b;
 RCLCPP_INFO(rclcpp::get_logger("rclcpp"), "Incoming request\na: %ld" " b: %ld",
                request->a, request->b);
 RCLCPP_INFO(rclcpp::get_logger("rclcpp"), "sending back response: [%ld]", (long int)
response->sum);
}
```

Request에 대해 실행되는 함수

Request에 대한 처리해 Response 넣기

Service - cpp

service_server.cpp

```cpp
int main(int argc, char **argv)
{
 rclcpp::init(argc, argv);

 std::shared_ptr<rclcpp::Node> node = rclcpp::Node::make_shared("add_two_ints_server");

 rclcpp::Service<ros2_interface::srv::ExCustomSrv>::SharedPtr service =
   node->create_service<ros2_interface::srv::ExCustomSrv>("add_two_ints", &add);

 RCLCPP_INFO(rclcpp::get_logger("rclcpp"), "Ready to add two ints.");
 rclcpp::spin(node);
 rclcpp::shutdown();
}
```

Node 생성

Service Server 생성

Service - cpp

service_client.cpp

```
$ cd ~/ros2_ws/src/ros2_cmake/src/
$ code service_client.cpp
```

```cpp
#include "rclcpp/rclcpp.hpp"
#include "ros2_interface/srv/ex_custom_srv.hpp"
#include <chrono>
#include <cstdlib>
#include <memory>
using namespace std::chrono_literals;
int main(int argc, char **argv)
{
 rclcpp::init(argc, argv);
 if (argc != 3) {
     RCLCPP_INFO(rclcpp::get_logger("rclcpp"), "usage: add_two_ints_client X Y");
     return 1;
 }
```

실행 인자를 받았는지 확인

Service - cpp

service_client.cpp

```
std::shared_ptr<rclcpp::Node> node = rclcpp::Node::make_shared("add_two_ints_client");
rclcpp::Client<ros2_interface::srv::ExCustomSrv>::SharedPtr client =
  node->create_client<ros2_interface::srv::ExCustomSrv>("add_two_ints");

auto request = std::make_shared<ros2_interface::srv::ExCustomSrv::Request>();
request->a = atoll(argv[1]);
request->b = atoll(argv[2]);
while (!client->wait_for_service(1s)) {
  if (!rclcpp::ok()) {
    RCLCPP_ERROR(rclcpp::get_logger("rclcpp"), "Interrupted while waiting for the service.
Exiting.");
    return 0;
  }
  RCLCPP_INFO(rclcpp::get_logger("rclcpp"), "service not available, waiting again...");
}
```

Node 생성

Service Client 생성

Request 생성

실행 인자 Request에 대입

1초 동안 Service 탐색

Service - cpp

service_client.cpp

```
auto result = client->async_send_request(request);
// Wait for the result.
if (rclcpp::spin_until_future_complete(node, result) ==
  rclcpp::FutureReturnCode::SUCCESS)
{
  RCLCPP_INFO(rclcpp::get_logger("rclcpp"), "Sum: %ld", result.get()->sum);
} else {
  RCLCPP_ERROR(rclcpp::get_logger("rclcpp"), "Failed to call service add_two_ints");
}
rclcpp::shutdown();
return 0;
}
```

Request 보내기

Result 출력

Service - cpp

CMakeList.txt

```
find_package(ros2_interface REQUIRED)
# service - server
add_executable(service_server src/service_server.cpp)
ament_target_dependencies(service_server rclcpp ros2_interface)
# service - client
add_executable(service_client src/service_client.cpp)
ament_target_dependencies(service_client rclcpp ros2_interface)
install(TARGETS
 # topic
 publisher
 subscriber
 # service
 service_server
 service_client
 DESTINATION lib/${PROJECT_NAME})
```

생성한 Service 찾기

Server, Client 추가

Server, Client 추가

Service - cpp

package.xml

생성한 ros2_interface 추가

```
<depend>ros2_interface</depend>
```
생성한 Service 의존성

Service - cpp

Build

```
$ cd ~/ros2_ws/
$ colcon build --packages-select ros2_cmake
$ source install/setup.bash
```

Service - cpp

실행

터미널 1 - Service Server	터미널 2 - Service Client
$ ros2 run ros2_cmake service_server	$ ros2 run ros2_cmake service_client 1 2

터미널 1 출력	터미널 2 출력
[INFO] [1735279654.719274102] [rclcpp]: Ready to add two ints. [INFO] [1735279660.526284176] [rclcpp]: Incoming request a: 1 b: 2 [INFO] [1735279660.526319516] [rclcpp]: sending back response: [3]	[INFO] [1735279660.526472607] [rclcpp]: Sum: 3

Service - python

service_server.py

```
$ cd ~/ros2_ws/src/ros2_python/ros2_python
$ code service_server.py
```

Service - python

service_server.py

```
from ros2_interface.srv import ExCustomSrv
import rclpy
from rclpy.node import Node
class CustomService(Node):
    def __init__(self):
        super().__init__('minimal_service')
        self.srv = self.create_service(ExCustomSrv, 'add_two_ints',
self.add_two_ints_callback)
    def add_two_ints_callback(self, request, response):
        response.sum = request.a + request.b
        self.get_logger().info('Incoming request\na: %d b: %d' % (request.a, request.b))
        return response
```

Request 처리

Service - python

service_server.py

```
def main():
    rclpy.init()
    custom_service = CustomService()
    rclpy.spin(custom_service)
    rclpy.shutdown()


if __name__ == '__main__':
    main()
```

Service - python

service_client.py

```
$ cd ~/ros2_ws/src/ros2_python/ros2_python
$ code service_client.py
```

Service - python

service_client.py

```
import sys
from ros2_interface.srv import ExCustomSrv
import rclpy
from rclpy.node import Node
class CustomClientAsync(Node):
    def __init__(self):
        super().__init__('minimal_client_async')
        self.cli = self.create_client(ExCustomSrv, 'add_two_ints')
        while not self.cli.wait_for_service(timeout_sec=1.0):
            self.get_logger().info('service not available, waiting again...')
        self.req = ExCustomSrv.Request()
    def send_request(self, a, b):
        self.req.a = a
        self.req.b = b
        return self.cli.call_async(self.req)
```

Request 함수

Service - python

service_client.py

```
def main():
    rclpy.init()
    custom_client = CustomClientAsync()
    future = custom_client.send_request(int(sys.argv[1]), int(sys.argv[2]))
    rclpy.spin_until_future_complete(custom_client, future)
    response = future.result()
    custom_client.get_logger().info(
        'Result of add_two_ints: for %d + %d = %d' % (int(sys.argv[1]), int(sys.argv[2]), response.sum))
    custom_client.destroy_node()
    rclpy.shutdown()
if __name__ == '__main__':
    main()
```

Request 보내기

Result 출력

Service - python

package.xml

- 생성한 ros2_interface 추가

```
<exec_depend>ros2_interface</exec_depend>
```

Service - python

setup.py

```
entry_points={
        'console_scripts': [
                            …
            'service_server = ros2_python.service_server:main',
            'service_client = ros2_python.service_client:main',
        ],
    },
```

Service - python

Build

```
$ cd ~/ros2_ws/
$ colcon build --packages-select ros2_python
$ source install/setup.bash
```

Service - python

Build

터미널 1	터미널 2
$ ros2 run ros2_python service_server	$ ros2 run ros2_python service_client 1 2

터미널 1 출력	터미널 2 출력
[INFO] [1735279881.776172112] [minimal_service]: Incoming request a: 1 b: 2	[INFO] [1735279881.788200266] [minimal_client_async]: Result of add_two_ints: for 1 + 2 = 3